

RepOrting Standards for Systematic Evidence Syntheses (ROSES)



ROSES: what is it?

- **ROSES = RepOrting standards for Systematic Evidence Syntheses** in Environmental management

What is provided?

- **Flow diagrams:** created by a team of researchers (CEE) experience in conducting systematic reviews in the environmental field, inspired by pre-existing tools from other disciplines (such as PRISMA in the medical field).
- **Forms:** specifically designed for **systematic reviews** and **systematic maps** in the field of environmental management developed by the *Collaboration for Environmental Evidence (CEE)*.

Transparency, rigor, and traceability: **key objectives**

Without transparent reporting, even well-designed reviews will fail to show their methodological strengths, undermining their utility in decision-making contexts (Pussegoda et al. 2017)

A ***persistent problem*** in Evidence synthesis:

- In “traditional”/ “non-systematic” reviews, the details of the **different steps and volumes** are only very rarely presented; they may sometimes be inferred, but are most often **completely omitted**.
Decisions are not documented.
- The review is therefore *not reproducible*!
- The reader cannot understand how the final result was obtained (lack of primary studies? high rate of inaccessible PDFs, etc.).

CEE Synthesis Assessment Tool: **CEESAT**



The CEESAT workshop group. From left: Jacqui Eales, Neal Haddaway, Ruth Garside, Nicola Randall, Barbara Livoreil, Andrew Pullin, Geoff Frampton, Christian Kohli and Biljana Macura.



1. THE REVIEW QUESTION

Rationale: A well-defined question (or hypothesis) is crucial for assessing the reliability of subsequent decisions on searching and screening for eligible studies, as well as forming the basis for critical appraisal of study conduct and for data extraction and synthesis.

1.1 Are the elements of the review question clear?

Tick whichever statement applies

- ☐ Gold: The review question or hypothesis is clearly stated and clearly defines key elements, (e.g. PICO, PECO, PO, PIT, etc.) correctly, such as the subject or population of interest, the intervention or exposure type, the comparator and valid measures of outcome.
- ☐ Green: The review question or hypothesis is clearly stated, and key-elements are mentioned although not formally defined in terms of PICO, PECO, PO, PIT, etc.
- ☐ Amber: The question or hypothesis is stated in broad terms but key-elements are unclear or poorly defined.
OR
Question or hypothesis not stated but problem or issue is stated such that a question can be inferred
- ☐ Red: A question, hypothesis or problem is not stated
OR
There is no stated objective to provide an answer to a question or test of a hypothesis.
OR
The article does not contain an evidence synthesis (e.g. primary research or descriptive overview)

CEESAT

2.1. Is there an a-priori method/protocol document?

- Gold: The review cites a separate a-priori protocol or documented pre-defined method containing details of proposed conduct of all review and synthesis stages (e.g. question, search, eligibility screening, critical appraisal, data extraction and synthesis)
NOTE: If the protocol is not specifically written for the review then the assessment should be green rather than gold.
AND
It is linked from the synthesis (e.g. as supplementary material or hosted on a separate website)
AND
It was publicly accessible prior to the conduct of the review
AND
It was submitted to an independent body for peer review and publication.
- Green: The review cites a separate accessible a-priori protocol or documented pre-defined method containing details of conduct of all review and synthesis stages (e.g. question, search, screening, critical appraisal, data extraction and synthesis)
AND
It is linked from the synthesis (e.g. as supplementary material or hosted on a separate website)
AND
It was publicly accessible prior to the conduct of the review.
- Amber: The review cites a separate accessible a-priori protocol or documented pre-defined method, but this does not contain all details of conduct of all review and synthesis stages or was not publicly accessible prior to the conduct of the review
OR
The review includes a defined methods section (not a-priori) listing the synthesis stages conducted and providing sufficient detail to enable the method to be replicated (therefore this standard is met only if all of criteria 3.1, 4.1, 6.1 & 7.1 are rated green or above).
- Red: There is no protocol and the review methods are not clearly defined in the methods section of the review or there are no methods reported.

4.3. Are eligibility decisions transparently reported?

Rationale: Listing all articles that were screened for eligibility and indicating whether each was included or excluded in data synthesis (usually as supplementary material), makes it clear whether potentially relevant studies have been omitted according to the eligibility criteria or were not captured by the search. Documenting the reasons for article exclusion at full text is essential for transparency.

- Gold: The total number of articles and number of unique articles found during the searches (after removal of duplicates) is presented
AND
The number excluded at each stage of the screening process is fully presented (e.g. in a flow diagram or table)
AND
Reasons for exclusion of each article/study considered at full-text are presented (e.g. in an appendix)
AND
A list of eligible (included) articles/studies is presented as a separate list or in tables (not just included in reference list).
- Green: The total number of articles found by the search and numbers excluded at each stage of the screening process is reported but some other aspects missing (e.g. number of unique articles or articles unobtainable)
AND
Reasons for exclusion of each article/study considered at full-text are presented (e.g. in an appendix)
AND
A list of eligible (included) articles/studies is presented as a separate list or in tables (not just included in reference list).
- Amber: The total number of articles found by the search and numbers excluded during the screening process is reported (or inferable) but some other aspects missing (e.g. number of unique articles or articles unobtainable or reasons for exclusion at full text)
AND
A list of eligible (included) articles/studies is presented as a separate list or in tables (not just included in reference list).
- Red: No to either or both of the amber criteria above

Standards and reporting

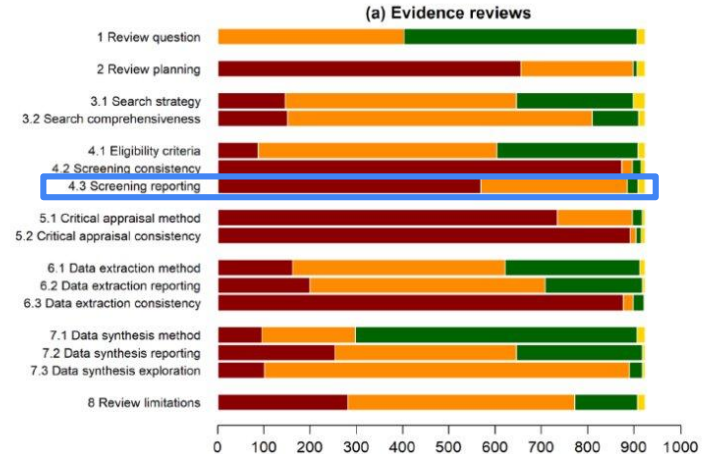
COMMENTARY

Open Access

Standards of conduct and reporting in evidence syntheses that could inform environmental policy and management decisions



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Source: Pullin *et al.* 2022. *Environ Evid*

Standards and reporting

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Systematic Review

Criteria

Systematic Reviews are reviews of evidence relevant to a clearly formulated question that uses systematic and explicit methods to identify, select and critically appraise relevant research, and to collect and analyse data from the studies that are included within the review. Authors should note that all Systematic Reviews published in *Environmental Evidence* will have been conducted according to the CEE process. Full guidelines and standards can be [accessed here](#) (a quick guide to the publication process is available in Section 1.1).

Note that Environmental Evidence expects all submitting authors to complete the relevant ROSES form as part of their submission to demonstrate that they have reported all relevant methodological details. Authors should also use the ROSES template for a flow diagram to report on the screening process. Templates for ROSES forms can be accessed here and the flow diagram here. ROSES forms and flow diagram should always be downloaded from the ROSES website as it contains most up-to-date templates. Please note the formatting of your submission should follow journal guidelines and not the ROSES template. The ROSES forms should be uploaded along with the submitted manuscript as a single-page supplementary file in a PDF format. Failure to do so could result in your manuscript being returned before review. In order to convert your completed ROSES form from a spreadsheet to a single-page PDF document, please scale the ROSES sheet to fit A4 landscape size.

For Systematic Reviews to be relevant to policy and practice they need to be as up-to-date as possible. Consequently, at the time of submission for publication, the search should normally be

Official journal of



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Evidence

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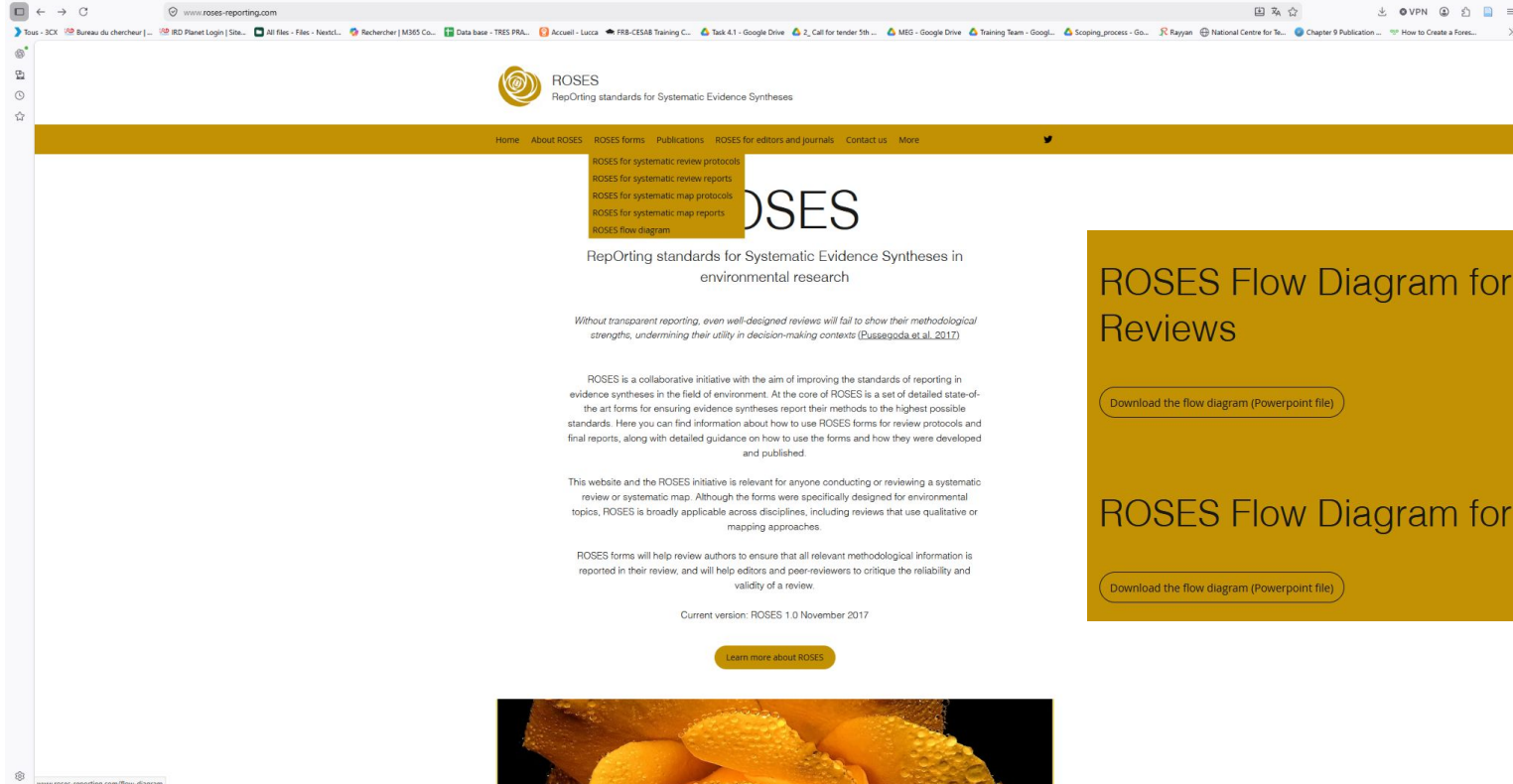
[Annual Journal Metrics](#)

ROSES flow diagrams

ROSES flow diagrams

- It is a **flowchart** that allows readers to quickly visualize the **entire systematic map or review process** and the reduction in study/article volumes through the different stages.
- It should be included as a **figure** in the manuscript of a systematic map/review.

ROSES flow diagrams



The screenshot shows the ROSES website with a navigation bar and a main content area. The navigation bar includes links for Home, About ROSES, ROSES forms, Publications, ROSES for editors and journals, Contact us, and More. The main content area features a large heading 'ROSES' and a subheading 'RepOrting standards for Systematic Evidence Syntheses in environmental research'. Below this, there is a quote: 'Without transparent reporting, even well-designed reviews will fail to show their methodological strengths, undermining their utility in decision-making contexts (Pussegoda et al. 2017)'. The text describes ROSES as a collaborative initiative aimed at improving the standards of reporting in evidence syntheses in the field of environment. It mentions that the core of ROSES is a set of detailed state-of-the-art forms for ensuring evidence syntheses report their methods to the highest possible standards. The website also states that ROSES is relevant for anyone conducting or reviewing a systematic review or systematic map, and that ROSES forms will help review authors to ensure that all relevant methodological information is reported in their review, and will help editors and peer-reviewers to critique the reliability and validity of a review. The current version is noted as ROSES 1.0 November 2017. A button labeled 'Learn more about ROSES' is visible at the bottom of the main content area. The footer of the website shows the URL 'www.roses-reporting.com/flow-diagram'.

ROSES
RepOrting standards for Systematic Evidence Syntheses

Home About ROSES ROSES forms Publications ROSES for editors and journals Contact us More

ROSES for systematic review protocols
ROSES for systematic review reports
ROSES for systematic map protocols
ROSES for systematic map reports
ROSES flow diagram

ROSES

RepOrting standards for Systematic Evidence Syntheses in environmental research

Without transparent reporting, even well-designed reviews will fail to show their methodological strengths, undermining their utility in decision-making contexts (Pussegoda et al. 2017)

ROSES is a collaborative initiative with the aim of improving the standards of reporting in evidence syntheses in the field of environment. At the core of ROSES is a set of detailed state-of-the-art forms for ensuring evidence syntheses report their methods to the highest possible standards. Here you can find information about how to use ROSES forms for review protocols and final reports, along with detailed guidance on how to use the forms and how they were developed and published.

This website and the ROSES initiative is relevant for anyone conducting or reviewing a systematic review or systematic map. Although the forms were specifically designed for environmental topics, ROSES is broadly applicable across disciplines, including reviews that use qualitative or mapping approaches.

ROSES forms will help review authors to ensure that all relevant methodological information is reported in their review, and will help editors and peer-reviewers to critique the reliability and validity of a review.

Current version: ROSES 1.0 November 2017

[Learn more about ROSES](#)

www.roses-reporting.com/flow-diagram

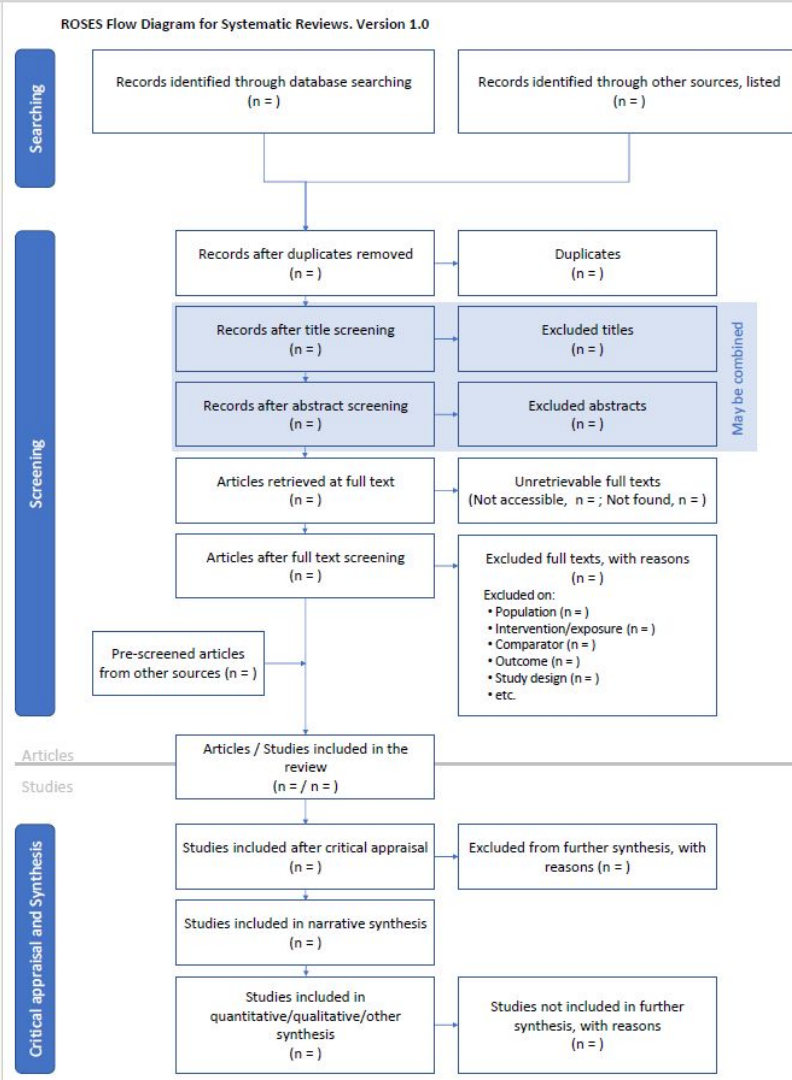
ROSES Flow Diagram for Systematic Reviews

[Download the flow diagram \(Powerpoint file\)](#)

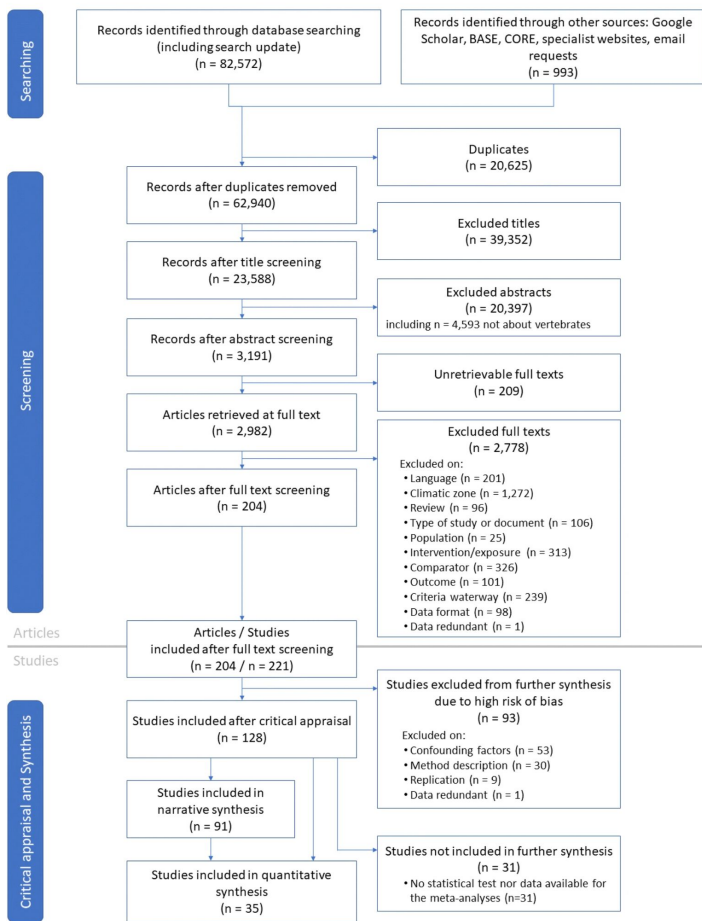
ROSES Flow Diagram for Systematic Maps

[Download the flow diagram \(Powerpoint file\)](#)

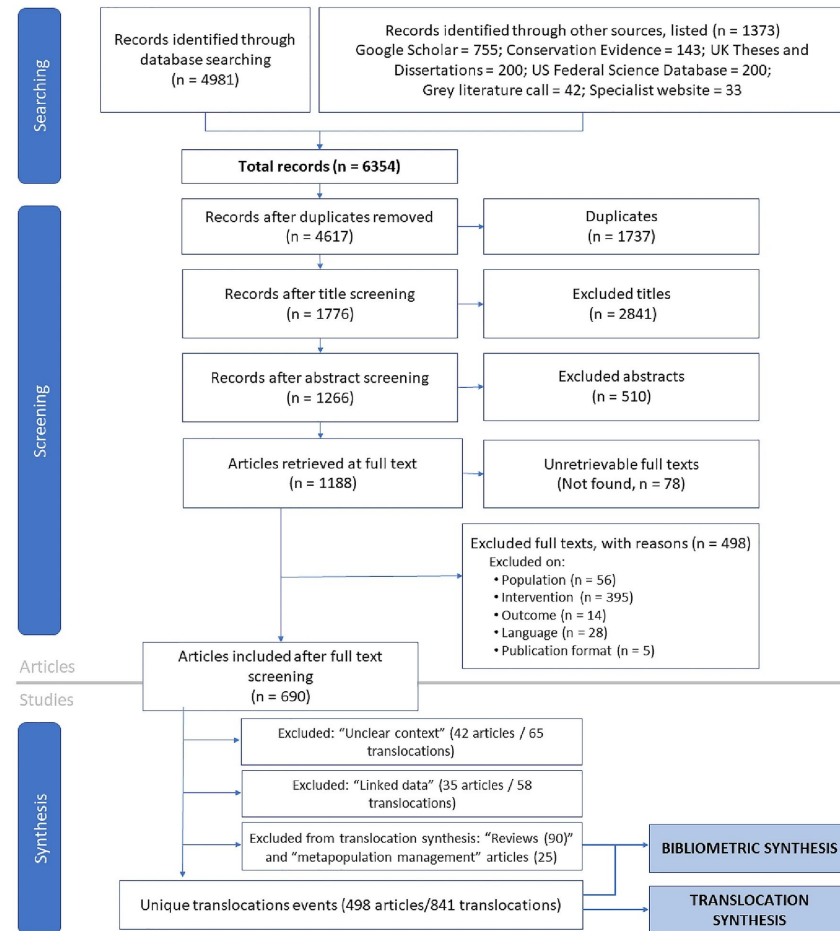
ROSES Flow Diagram for Systematic Reviews



ROSES Flow Diagram for Systematic Reviews. Version 1.0



ROSES Flow Diagram for Systematic Maps. Version 1.0



Ensuring transparency and traceability

biblio internal id	biblio authors	biblio title	biblio container	biblio abstract	biblio year	biblio doi	biblio woskey	duplicate check	screen title decision	Articles qualified for abstract screening	screen abstract decision	pdf retrieved	file name	submitted to full text screening	screen fulltext language	screen fulltext pico	screen fulltext pico justify	screen fulltext decision	screen fulltext decision justify
15	Beschta, R.; I	Can large carnivores change stream	ECOHYDROL	Large carniv	2019	10.1002/eco	WOS:0004544	No: retained for screening	Y	Yes: retained on title	Y	YES	Beschta, et al	Yes: retained on abstract	en	N	o	N	pico exclusion
19	Bakker, ES; S	Trophic rewinding: impact on ecosystems	PHILOSOPHIC	no abstract	2018	10.1098/rstb	WOS:0004449	No: retained for screening	Y	Yes: retained on title	No-Abstract	YES	Bakker, and	Yes: retained by default - no	en	N	i	N	pico exclusion
27	Walker, LE; D	Population responses of common ravens	ECOLOGICAL	Top predator	2018	10.1002/ecs	WOS:0004514	No: retained for screening	Y	Yes: retained on title	Y	YES	Walker, et al	Yes: retained on abstract	en	N	i	N	pico exclusion
31	Kuiper, TR; D	Demography and social dynamics of	JOURNAL OF	1. Given them	2018	10.1111/1365	WOS:0004447	No: retained for screening	Y	Yes: retained on title	Y	YES	Kuiper, et al	Yes: retained on abstract	en	Y	NA	Y	NA
32	Buk, KG; van	Conservation of severely fragmented	BIODIVERSITY	We document	2018	10.1007/s105	WOS:000445	No: retained for screening	Y	Yes: retained on title	Y	YES	Buk, et al	Yes: retained on abstract	en	Y	NA	Y	NA
34	Boyce, MS	Wolves for Yellowstone: dynamics of	JOURNAL OF	The reintrod	2018	10.1093/jma	WOS:000448	No: retained for screening	Y	Yes: retained on title	Y	YES	Boyce, 2018	Yes: retained on abstract	en	N	i	N	pico exclusion
39	Sagua, M; Fj	Inferring the origin and genetic div	MAMMAL RE	The Eurasian	2018	10.1007/s133	WOS:000444	No: retained for screening	Y	Yes: retained on title	Y	YES	Sagua, et al	Yes: retained on abstract	en	N	i	N	pico exclusion
47	Watkins, GE	Success criteria not met, but valuab	WILDLIFE RES	Context. Fau	2018	10.1071/WR	WOS:000444	No: retained for screening	Y	Yes: retained on title	Y	YES	Watkins, et al	Yes: retained on abstract	en	Y	NA	Y	NA
50	Painter, LE; B	Aspen recruitment in the Yellowstone	ECOSPHERE	Quaking aspe	2018	10.1002/ecs	WOS:000444	No: retained for screening	Y	Yes: retained on title	Y	YES	Painter, et al	Yes: retained on abstract	en	N	o	N	pico exclusion
51	Zheng, LM; H	Habitat Evaluation for Reintroduced	PAKISTAN J	Crested ibis	2018	10.17582/jou	WOS:000447	No: retained for screening	Y	Yes: retained on title	Y	YES	Zheng, et al	Yes: retained on abstract	en	Y	NA	Y	NA
53	Muller, L; M	Genetic structure in Elk persists after	JOURNAL OF	Elk (Capreolus	2018	10.1002/jwn	WOS:000438	No: retained for screening	Y	Yes: retained on title	Y	YES	Muller, et al	Yes: retained on abstract	en	Y	NA	Y	NA
55	Heurich, M	Illegal hunting as a major driver of	BIOLOGICAL	(Large carniv	2018	10.1016/j.bi	WOS:000439	No: retained for screening	Y	Yes: retained on title	Y	YES	Heurich, et al	Yes: retained on abstract	en	Y	NA	Y	NA
63	Kenup, CF; S	Walking on their own legs: unassiste	ORYX	Reintroducti	2018	10.1017/S003	WOS:000450	No: retained for screening	Y	Yes: retained on title	Y	YES	Kenup, et al	Yes: retained on abstract	en	Y	NA	Y	NA
65	Vasile, M	The Vulnerable Bison: Practices and	CONSERVATI	This paper di	2018	10.4103/cs.c	WOS:000445	No: retained for screening	Y	Yes: retained on title	Y	YES	Vasile, 2018	Yes: retained on abstract	en	N	p	N	pico exclusion
67	Elbroch, JM	Multiple anthropogenic intervention	ECOLOGICAL	Humans are	2018	10.1002/ecs	WOS:000440	No: retained for screening	Y	Yes: retained on title	Y	YES	Elbroch, et al	Yes: retained on abstract	en	N	i	N	pico exclusion
68	Miranda, EB	Reintroducing apex predators: the	ROYAL SOCI	no abstract	2018	10.1098/rsos	WOS:000440	No: retained for screening	Y	Yes: retained on title	No-Abstract	YES	Miranda, 2018	Yes: retained by default - no	en	N	p	N	pico exclusion
69	Smith, TN; K	Diet Composition and Selection of	AMERICAN J	Since elc (Cae	2018	10.1674/0003	WOS:000438	No: retained for screening	Y	Yes: retained on title	Y	YES	Smith, et al	Yes: retained on abstract	en	Y	NA	Y	NA
71	Rodriguez-R	The restoration of the endangered	PEERJ	The transloc	2018	10.7717/peer	WOS:000435	No: retained for screening	Y	Yes: retained on title	Y	YES	Rodriguez-R	Yes: retained on abstract	en	Y	NA	Y	NA
73	Cortez, MV; F	Effect of antipredator training on sp	ACTA ORNITH	The Greater f	2018	10.3161/0001	WOS:000434	No: retained for screening	Y	Yes: retained on title	Y	YES	Cortez, et al	Yes: retained on abstract	en	Y	NA	Y	NA
75	Renan, S; P	Fission-fusion social structure of a	BIOLOGICAL	(In a reintrod	2018	10.1016/j.bi	WOS:000434	No: retained for screening	Y	Yes: retained on title	Y	YES	Renan, et al	Yes: retained on abstract	en	Y	NA	Y	NA
77	Woodruff, SP	Characteristics of Winter Wolf Kill	S JOURNAL OF	Understanding	2018	10.3996/032	WOS:000434	No: retained for screening	Y	Yes: retained on title	Y	YES	Woodruff, et al	Yes: retained on abstract	en	Y	NA	Y	NA
87	Hale, SL; K	Ecosystem-level effects of keystone	RESTORATION	The keystone	2018	10.1111/rec	WOS:000434	No: retained for screening	Y	Yes: retained on title	Y	YES	Hale, and K	Yes: retained on abstract	en	N	i	N	pico exclusion
96	Makin, DF; C	Changes in feeding behavior and pa	JOURNAL OF	Top-order ca	2018	10.1093/jma	WOS:000429	No: retained for screening	Y	Yes: retained on title	Y	YES	Makin, et al	Yes: retained on abstract	en	N	i	N	pico exclusion
99	Codron, D; R	Meso-carnivore niche expansion in	AFRICAN JOL	Apex predat	2018	10.3957/056	WOS:000432	No: retained for screening	Y	Yes: retained on title	Y	YES	Codron, et al	Yes: retained on abstract	en	N	o	N	pico exclusion
101	Abell, T; Cau	A gleam of hope for the critically en	AQUATIC CO	Results of th	2018	10.1002/aq	WOS:000429	No: retained for screening	Y	Yes: retained on title	Y	YES	Abell, et al	Yes: retained on abstract	en	Y	NA	Y	NA
109	Jolly, C; K	Out of the frying pan: Reintroduction	AUSTRALIAN	ECOInvasive spe	2018	10.1111/aec	WOS:000428	No: retained for screening	Y	Yes: retained on title	Y	YES	Jolly, et al	Yes: retained on abstract	en	Y	NA	Y	NA
110	Beschta, R; I	Trophic cascades at multiple spatial	FOREST ECO	Throughout e	2018	10.1016/j.for	WOS:000427	No: retained for screening	Y	Yes: retained on title	Y	YES	Beschta, et al	Yes: retained on abstract	en	N	o	N	pico exclusion
112	Zarrouk, A; R	Reintroduction of the highly endang	ESTUARINE	(Patella ferrug	2018	10.1016/j.ec	WOS:000426	No: retained for screening	Y	Yes: retained on title	Y	YES	Zarrouk, et al	Yes: retained on abstract	en	Y	NA	Y	NA
113	Stover, H; N	Transplanting Foreign Non-Native	ECOLOGICAL	Disturbed are	2018	10.3368/er	WOS:000456	No: retained for screening	Y	Yes: retained on title	Y	YES	Stover, et al	Yes: retained on abstract	en	Y	NA	Y	NA
116	Scaffaro, M	Beaver Habitat Selection for 24 Yr	SIPANGELAND	Beavers (Cast	2018	10.1016/j.w	WOS:000426	No: retained for screening	Y	Yes: retained on title	Y	YES	Scaffaro, et al	Yes: retained on abstract	en	Y	NA	Y	NA
128	Valko, O; T	Cultural heritage and biodiversity c	NATURE CON	Understanding	2018	10.3897/nat	WOS:000419	No: retained for screening	Y	Yes: retained on title	Y	YES	Valko, et al	Yes: retained on abstract	en	Y	NA	Y	NA
131	Bakka, SV; K	The story of the creation and moni	ALL-RUSSIAN	The article su	2018	10.1088/1755	WOS:000454	No: retained for screening	Y	Yes: retained on title	Y	YES	Bakka, et al	Yes: retained on abstract	en	Y	NA	Y	NA
143	Vogel, JT; S	The foraging ecology of reintroduc	WILDLIFE BIO	Restoring lar	2018	10.2981/wlb	WOS:000439	No: retained for screening	Y	Yes: retained on title	Y	YES	Vogel, et al	Yes: retained on abstract	en	Y	NA	Y	NA
146	Robinson, N	Bandicoots return to Boodeere: in	WILDLIFE RES	Context. Reir	2018	10.1071/WR	WOS:000431	No: retained for screening	Y	Yes: retained on title	Y	YES	Robinson, et al	Yes: retained on abstract	en	Y	NA	Y	NA
147	Ismaili, B; D	Getting the dietary knowledge to re	WILDLIFE BIO	Atlas deer Ce	2018	10.2981/wlb	WOS:000430	No: retained for screening	Y	Yes: retained on title	Y	YES	Ismaili, et al	Yes: retained on abstract	en	Y	NA	Y	NA
153	Leung, FTY; Y	MECHANICAL PULL-OUT CAPACITY	A JOURNAL OF	Ecological re	2018	10.26525/jfs	WOS:000422	No: retained for screening	Y	Yes: retained on title	Y	YES	Leung, et al	Yes: retained on abstract	en	N	i	N	pico exclusion
154	Brzozko, E; J	Rare plant translocation between	RESTORATION	How carried o	2018	10.1111/rec	WOS:000419	No: retained for screening	Y	Yes: retained on title	Y	YES	Brzozko, et al	Yes: retained on abstract	en	Y	NA	Y	NA
159	Wu, J; Zhu, J	Breeding Ecology of a Reintroduc	PAKISTAN JOT	The crested i	2017	10.17582/jou	WOS:000424	No: retained for screening	Y	Yes: retained on title	Y	YES	Wu, et al	Yes: retained on abstract	en	Y	NA	Y	NA
160	Kovacic, T; P	Adaptation of cultured mussel Mytil	ACTA ADRIAT	The ability of	2017	no DOI	WOS:000423	No: retained for screening	Y	Yes: retained on title	Y	YES	Kovacic, et al	Yes: retained on abstract	en	N	i	N	pico exclusion
161	Cremona, T; H	High mortality and small population	ANIMAL CON	Northern Aus	2017	10.1111/acv	WOS:000423	No: retained for screening	Y	Yes: retained on title	Y	YES	Cremona, et al	Yes: retained on abstract	en	Y	NA	Y	NA
167	Cremona, T; J	Avoiding the last supper: parentage	CONSERVATI	Invasive spec	2017	10.1007/s105	WOS:000414	No: retained for screening	Y	Yes: retained on title	Y	YES	Cremona, et al	Yes: retained on abstract	en	Y	NA	Y	NA
171	Sarkar, MS; K	Assessment of fine-scale resource	PEERJ	Background.	2017	10.7717/peer	WOS:000414	No: retained for screening	Y	Yes: retained on title	Y	YES	Sarkar, et al	Yes: retained on abstract	en	Y	NA	Y	NA
176	McClure, CJW	Successful enhancement of Ridgway	CONDOR	Survival and	2017	10.1650/CON	WOS:000415	No: retained for screening	Y	Yes: retained on title	Y	YES	McClure, et al	Yes: retained on abstract	en	Y	NA	Y	NA
180	Benjamin-Fin	Conservation implications of wildlif	GLOBAL ECO	Wildlife trans	2017	10.1016/j.g	WOS:000426	No: retained for screening	Y	Yes: retained on title	Y	YES	Benjamin-Fin	Yes: retained on abstract	en	N	p	N	pico exclusion
183	Zamboni, T; A	review of a multispecies reintrod	PERSPECTIVE	Libera Nature	2017	10.1016/j.p	WOS:000418	No: retained for screening	Y	Yes: retained on title	Y	YES	Zamboni, et al	Yes: retained on abstract	en	Y	NA	Y	NA
184	Fernandez, F	Rewilding the Atlantic Forest: Resto	PERSPECTIVE	The loss or re	2017	10.1016/j.p	WOS:000418	No: retained for screening	Y	Yes: retained on title	Y	YES	Fernandez, et al	Yes: retained on abstract	en	Y	NA	Y	NA

Ensuring transparency and traceability

Minimum requirement
: provide full text exclusion
decisions!

pdf retrieved	file name	submitted to full text screening	fulltext kappa sample (%)	fulltext kappa decision	fulltext kappa decision PC	screen fulltext language	screen fulltext pico	screen fulltext pico justif	screen fulltext decision	screen fulltext decision justify
YES	Beschta et al 2019	Yes: retained on abstract	No	NA	NA	en	N	o	N	pico_exclusion
YES	Bakker and Svenning 2018	Yes: retained by default - no abstract	No	NA	NA	en	N	i	N	pico_exclusion
YES	Walker et al 2018	Yes: retained on abstract	No	NA	NA	en	N	i	N	pico_exclusion
YES	Boyce 2018	Yes: retained on abstract	No	NA	NA	en	N	i	N	pico_exclusion
YES	Sagua et al 2018	Yes: retained on abstract	No	NA	NA	en	N	i	N	pico_exclusion
YES	Painter et al 2018	Yes: retained on abstract	No	NA	NA	en	N	o	N	pico_exclusion
YES	Vasile 2018	Yes: retained on abstract	No	NA	NA	en	N	p	N	pico_exclusion
YES	Elbroch et al 2018	Yes: retained on abstract	No	NA	NA	en	N	i	N	pico_exclusion
YES	Miranda 2018	Yes: retained by default - no abstract	Yes	N	N	en	N	p	N	pico_exclusion
YES	Hale and Koprowski 2018	Yes: retained on abstract	No	NA	NA	en	N	i	N	pico_exclusion
YES	Makin et al 2018	Yes: retained on abstract	No	NA	NA	en	N	i	N	pico_exclusion
YES	Codron et al 2018	Yes: retained on abstract	No	NA	NA	en	N	o	N	pico_exclusion
YES	Beschta et al 2018	Yes: retained on abstract	No	NA	NA	en	N	o	N	pico_exclusion
YES	Leung et al 2018	Yes: retained on abstract	No	NA	NA	en	N	i	N	pico_exclusion
YES	Kovacic et al 2017	Yes: retained on abstract	No	NA	NA	en	N	i	N	pico_exclusion
YES	Benjamin-Fink and Reilly 2017	Yes: retained on abstract	No	NA	NA	en	N	p	N	pico_exclusion
YES	Elbroch et al 2017	Yes: retained on abstract	No	NA	NA	en	N	i	N	pico_exclusion
YES	O'Connor and Goodall 2017	Yes: retained on abstract	No	NA	NA	en	N	p	N	pico_exclusion
YES	Sujii et al 2017	Yes: retained on abstract	No	NA	NA	en	N	i	N	pico_exclusion
YES	Jimenez et al 2017	Yes: retained on abstract	No	NA	NA	en	N	i	N	pico_exclusion
YES	Baker et al 2017	Yes: retained on abstract	No	NA	NA	en	N	i	N	pico_exclusion
YES	Kalle et al 2017	Yes: retained on abstract	No	NA	NA	en	N	i	N	pico_exclusion
YES	Cuadrado et al 2017	Yes: retained on abstract	No	NA	NA	other	N	NA	N	language_exclusion
YES	Phillips 2015	Yes: retained on abstract	No	NA	NA	en	N	p	N	pico_exclusion
YES	O'Connor 2017	Yes: retained on abstract	No	NA	NA	en	N	i	N	pico_exclusion
YES	Davies et al 2016	Yes: retained on abstract	No	NA	NA	en	N	i	N	pico_exclusion
YES	Beschta and Ripple 2016	Yes: retained on abstract	No	NA	NA	en	N	i	N	pico_exclusion
YES	Miller et al 2016	Yes: retained on abstract	No	NA	NA	en	N	i	N	pico_exclusion
YES	Ziolkowska et al 2016	Yes: retained on abstract	No	NA	NA	en	N	o	N	pico_exclusion
YES	Bleyhl et al 2015	Yes: retained on abstract	No	NA	NA	en	N	i	N	pico_exclusion
YES	Ripple et al 2015	Yes: retained on abstract	No	NA	NA	en	N	i	N	pico_exclusion
YES	Elboch et al 2015	Yes: retained on abstract	No	NA	NA	en	N	i	N	pico_exclusion
YES	Linnane et al 2015	Yes: retained on abstract	No	NA	NA	en	N	i	N	pico_exclusion
YES	Painter et al 2015	Yes: retained on abstract	No	NA	NA	en	N	i	N	pico_exclusion
YES	Loureiro et al 2015	Yes: retained on abstract	No	NA	NA	en	N	p	N	pico_exclusion
YES	Beschta and Ripple 2014	Yes: retained on abstract	No	NA	NA	en	N	i	N	pico_exclusion
YES	Popp et al 2014	Yes: retained on abstract	No	NA	NA	en	N	i	N	pico_exclusion
Yes	WHITTINGTON-JONES et al 2014	Yes: retained on abstract	No	NA	NA	en	N	i	N	pico_exclusion
YES	Painter et al 2014	Yes: retained on abstract	No	NA	NA	en	N	i	N	pico_exclusion
YES	Weise et al 2014	Yes: retained on abstract	No	NA	NA	en	N	i	N	pico_exclusion
YES	Christianson and Creel 2014	Yes: retained on abstract	No	NA	NA	en	N	i	N	pico_exclusion
YES	Wilson et al 2014	Yes: retained on abstract	No	NA	NA	en	N	i	N	pico_exclusion
YES	Runyon et al 2014	Yes: retained on abstract	No	NA	NA	en	N	i	N	pico_exclusion
YES	Strong et al 2013	Yes: retained on abstract	No	NA	NA	en	N	i	N	pico_exclusion
Yes	Kohl et al 2013	Yes: retained on abstract	No	NA	NA	en	N	i	N	pico_exclusion
YES	Herrero et al 2013	Yes: retained on abstract	No	NA	NA	en	N	i	N	pico_exclusion
YES	Miller et al 2013	Yes: retained on abstract	No	NA	NA	en	N	i	N	pico_exclusion
YES	Samelius et al 2013	Yes: retained on abstract	No	NA	NA	en	N	i	N	pico_exclusion
YES	Runge 2013	Yes: retained on abstract	No	NA	NA	en	N	i	N	pico_exclusion
YES	Hugron et al 2013	Yes: retained on abstract	No	NA	NA	en	N	i	N	pico_exclusion
YES	Kauffman et al 2013	Yes: retained by default - no abstract	No	NA	NA	en	N	i	N	pico_exclusion
YES	Vallone et al 2013	Yes: retained on abstract	No	NA	NA	en	N	p	N	pico_exclusion
YES	Marshall et al 2013	Yes: retained on abstract	No	NA	NA	en	N	o	N	pico_exclusion
Yes	Louw et al 2012	Yes: retained on abstract	No	NA	NA	en	N	i	N	pico_exclusion

ROSES forms

ROSES forms

Why is it important?

- **Facilitates manuscript writing for authors:** *helps ensure that the all correct information has been included with the appropriate level of detail adhering to Collaboration for Environmental Evidence guidelines*
- **Supports the writing process:** avoid manuscript revisions or rejections
- **Serves as a “checklist”:** really useful if publishing in Environmental Evidence Journal (EEJ)
- **Serves as a guarantee of transparency and rigor.**

ROSES forms

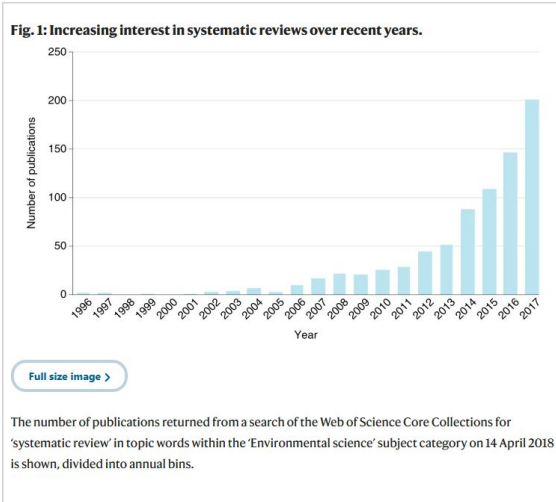
Comment | Published: 30 May 2018

The role of reporting standards in producing robust literature reviews

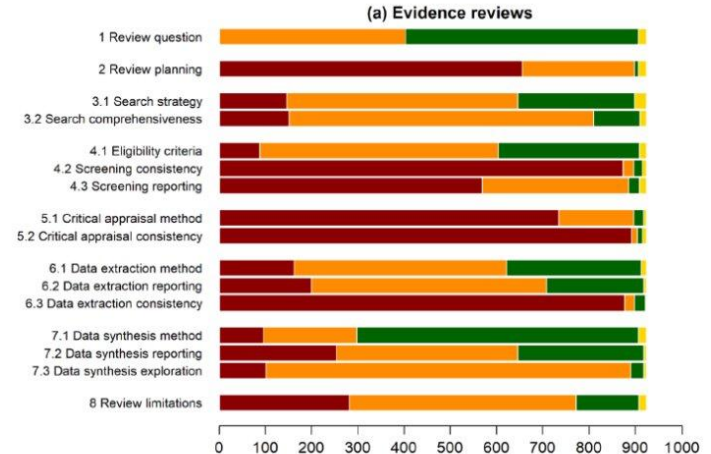
Neal Robert Haddaway  & Biljana Macura

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Increasing interest but poor consideration for reporting and transparency.



Source: Pullin *et al.* 2022. *Environ Evid*

Source: Haddaway & Macura. 2018. *Nature climate change*.

ROSES forms for systematic maps



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Type of review	Type of review	Select one of the following types of review: systematic map, systematic map update, systematic map amendment	See CEE Guidance on systematic mapping [1], and on amendments and updates [2]	Meta-data	systematic map	
Authors' contacts	Authors' contacts	The full names, institutional addresses and email addresses for all authors must be provided.		Checklist	Yes	
Abstract	Structured summary	The abstract of the manuscript must not exceed 500 words and must be structured into separate sections: Background, the context and purpose of the review, including the review questions; Methods, how the review was performed (specifically mention search strategy, inclusion criteria, critical appraisal (optional), meta-data extraction and coding, and narrative synthesis); Results, the main findings, including results of search and assessment of evidence base; Conclusions, brief summary and potential implications for policy/management and research.	A theory of change and/or conceptual model should be presented that links the intervention or exposure to the outcome.	Checklist	Yes	
Background	Background	Describe the rationale for the review in the context of what is already known. Reviews must indicate why this study was necessary and what it aims to contribute to the field. The actual role of stakeholders throughout the review process (e.g. in the formulation of the question) must be described and explained (using a broad definition of 'stakeholder', including e.g. researchers, funders and other decision-makers; see [3])		Checklist	Yes	
Stakeholder engagement	Stakeholder engagement		The primary question is the main question of the review. The secondary questions are usually linked to sources of heterogeneity (effect modifiers).	Checklist	Yes	
Objective of the review	Objective	Describe the primary question and secondary questions (when applicable).		Checklist	Yes	
Definition of the question components	Definition of the question components	Provide reference to the question key elements, e.g. population(s), intervention(s)/exposure(s), comparator(s), and outcome(s).	For other question types see [4,5]	Meta-data	Population (P): All species (fauna, flora, fungi) of wild or captive source. Intervention (I): All conservation-based translocations (reintroductions, supplementations (reinforcements) and introductions (assisted migrations)). Comparator (C): Studies will not be required <i>stricto sensu</i> to have a comparator. Outcomes (O): All relevant outcomes related to the translocated population e.g. survival, demography, space use, behavior, reproduction.	Refer to Table 1 of manuscript for full breakdown
Methods	Protocol	Provide citation, DOI or open access link to published protocol.	The protocol should be peer-reviewed and	Meta-data	https://doi.org/10.1186/s12750-020-00199-4	
Searches	Search strategy	Describe any ways in which the final methods of the review deviate from the protocol. Detail the search strategy used, including: database names accessed, search string, languages, bibliographic databases, and web-based search engines.		Checklist	Yes	
Search string	Search string	Provide Boolean-style full search string and state the platform for which the string is formatted (e.g. Web of Science format)		Meta-data	TS = ("protected areas" OR "natur* reserves" OR "national parkS" OR "regional parkS" OR "national reserveS" OR "biological reserveS" OR "regional reserveS") AND TS = ("assisted colonization" OR "assisted population migration" OR "assisted migration" OR "assisted gene flow" OR "managed relocation" OR "transloc*" OR "reintrodu*" OR "reintro*" OR "assisted range expansion" OR "assisted long-distance migrationS")	
Languages - bibliographic	Languages - bibliographic	List languages used in bibliographic database searches		Meta-data	English and French	
Languages - grey literature	Languages - grey literature	List languages used in organisational website searches and web-		Meta-data	English and French	
Bibliographic databases	Bibliographic databases	Provide the number of bibliographic databases searched		Meta-data	Two	Web of science (core collection); Scopus
Web-based search engines	Web-based search engines	Provide the number of web-based search engines searched		Meta-data	Four	Google Scholar, Conservation Evidence, UK Theses and Dissertations, US Federal
Organisational websites	Organisational websites	Provide the number of organisational websites searched		Meta-data	Five	1) US Fish and wildlife service (https://www.fws.gov/) 2) Office National de la Chasse et de la Faune Sauvage (http://www.oncfs.gouv.fr/) 3) IUCN Conservation Planning Specialist Group's document library (http://www.cpsg.org/document-repository) 4) Association of Zoos & Aquariums (https://www.aza.org/), Including the European Association (https://www.eaza.net/) 5) Rewilding Europe (https://rewildingeurope.com/)

Source: Langridge et al. 2021. *Environ Evid*